

Hybrid Few-Shot Anomaly Detection System

1. Introduction

Anomaly detection in time-series data plays a crucial role in many real-world applications such as financial fraud detection, industrial monitoring, healthcare systems, and cybersecurity. However, anomalies are rare and labeled data is often limited, making traditional machine learning approaches less effective. To address this issue, this report proposes a Hybrid Few-Shot Anomaly Detection System that can learn from a very small amount of data and still achieve high performance.

2. Problem Statement

Detecting anomalies in time-series data presents several challenges:

- Anomalies are rare and difficult to identify
- Lack of labeled data and high labeling cost
- Complex patterns including trend, seasonality, and noise
- Traditional models require large datasets

Therefore, there is a need for a model that is data-efficient, adaptive, and capable of handling complex time-series patterns.

3. Proposed System Overview

The proposed solution is a hybrid model that combines multiple deep learning techniques to improve anomaly detection performance. The system is designed to work effectively even with limited labeled data using few-shot learning.

4. System Architecture

The architecture consists of the following components:

4.1 LSTM (Long Short-Term Memory)

LSTM is used to capture temporal dependencies and sequential patterns in time-series data. It helps in understanding trends and patterns over time.

4.2 Transformer

The Transformer component captures long-range dependencies and complex relationships within the data, which traditional models may fail to detect.

4.3 Siamese Network

A Siamese Network is used for few-shot learning. It learns similarity between data samples and helps distinguish between normal and anomalous patterns.

4.4 Distance-Based Scoring

A distance-based method is used to calculate how far a data point deviates from normal behavior. If the distance exceeds a threshold, it is classified as an anomaly.

5. Learning Strategy

The model follows a two-stage learning process:

5.1 Self-Supervised Pretraining

- Masked time-series modeling
- Contrastive learning

This stage allows the model to learn meaningful feature representations without requiring labeled data.

5.2 Few-Shot Fine-Tuning

- Uses a small number of normal samples
- Adapts the model to the target domain

This significantly reduces the dependency on large labeled datasets.

6. Key Features

- Efficient with limited labeled data
- Handles complex and noisy time-series data
- Supports real-time anomaly detection
- Detects both seen and unseen anomalies

7. Explainability and Adaptation

To improve trust and usability, the system includes:

- Attention visualization to highlight important time steps
- Adaptive thresholding for better anomaly detection
- Memory bank to recognize unseen anomaly patterns

8. Expected Results

The proposed system is expected to:

- Achieve higher accuracy (AUC and F1-score)
- Perform well in small-data scenarios
- Provide robust anomaly detection in real-world environments

9. Applications

This system can be applied in:

- Financial fraud detection
- Industrial IoT monitoring
- Healthcare monitoring systems
- Network security and intrusion detection

10. Conclusion

The Hybrid Few-Shot Anomaly Detection System provides an efficient and scalable solution for detecting anomalies in time-series data. By combining LSTM, Transformer, and Siamese networks with self-supervised learning, the system overcomes the limitations of traditional methods and performs effectively in real-world scenarios with limited data.

11. Future Work

Future improvements may include:

- Deployment on edge devices
- Integration with real-time monitoring systems
- Enhancing model interpretability
- Expanding to multimodal anomaly detection